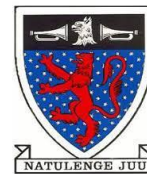


STAREHE BOYS' 2025 PREDICTION



Kenya Certificate of Secondary Education

ANTICIPATED KCSE EXAM



231/2

Biology (Theory)

Paper 2

Time: 2 Hours

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

- Write your name and Index Number in the spaces provided above.
- This paper consists of **two** sections. Section **A** and section **B**.
- Answer **ALL** questions in section **A** in the spaces provided. In section **B** answer question **6** (compulsory) and either question **7** or **8** in the spaces provided after question 8

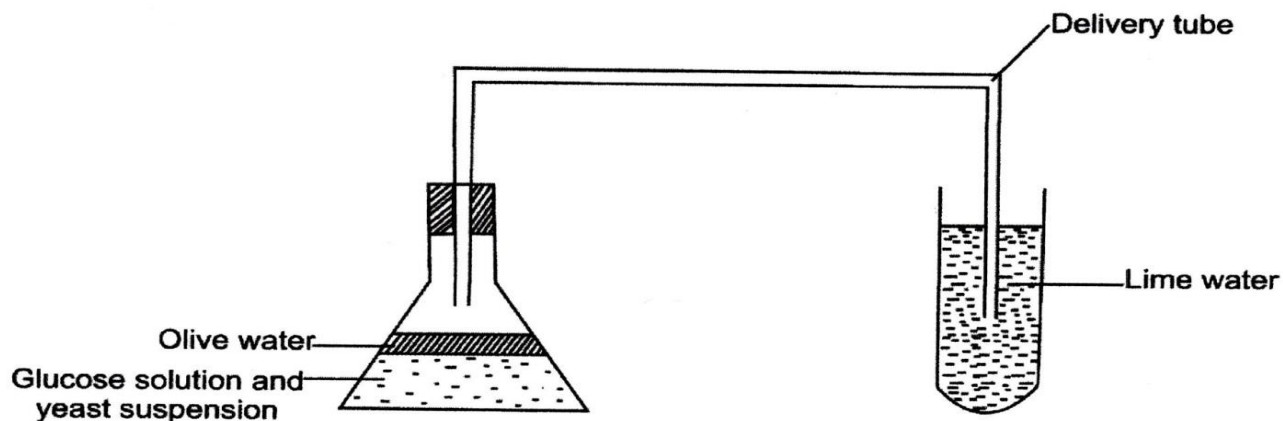
FOR EXAMINERS USE ONLY.

Section	Question	Maximum score	Candidates score
A	1	8	
	2	8	
	3	8	
	4	8	
	5	8	
B	6	20	
	7/8	20	
	Total score	80	

SECTION A – (40 MARKS)

Answer All Questions In This Section In The Spaces Provided.

1. The set up below was used to investigate a certain process.



Glucose solution was boiled and olive oil added on top. It was then allowed to cool before the yeast suspension was added.

- a) Name the biological process being investigated. (1 mark)

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- b) Why was the glucose solution boiled? (1 mark)

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c) What is the importance of cooling the glucose solution before adding the yeast? **(1 mark)**

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d) What is the role of olive oil in the experiment? **(1 mark)**

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e) What will be observed in the lime water at the end of the experiment? **(1 mark)**

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f) Suggest what would be observed if a thermometer reading 25°C was inserted in the glucose solution with the yeast suspension **(2 marks)**

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g) Suggest a control experiment for this set up. **(1 mark)**

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2. In maize the gene for Green colour is dominant to the gene for white colour. A pure breeding maize plant with green grains was crossed with heterozygous plant.

a) (i) Using letter G to represent the gene for green colour, work out the genotypic ratio of the offspring. **(5 marks)**

(ii) State the phenotype of the offspring. (1 mark)

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b) What is genetic engineering? (1 mark)

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c) What is meant by hybrid vigour? (1 mark)

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3. (a) Briefly discuss how the following gives evidences on specification:

(i) Cell biology (2 marks)

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(ii) Comparative embryology (2 marks)

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(b) State one limitation of using fossil records as an evidence of organic evolution. (1 mark)

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(c) (i) What is meant by the term vestigial structure. (1 mark)

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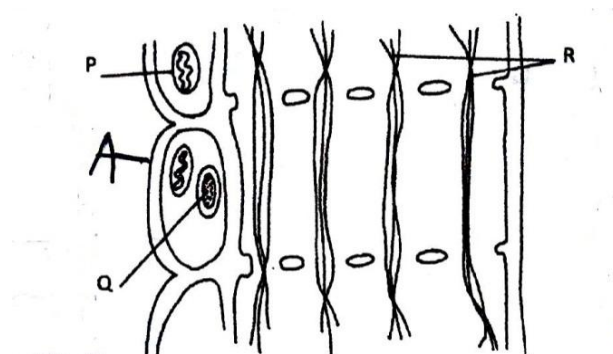
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(ii) Name **two** examples of vestigial structures in man. (2 marks)

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4. The diagram below shows a tissue in a plant.



a) (i) **Identify** the tissue shown in the diagram above. (1 mark)

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(ii) State the function of the tissue shown above. (1 mark)

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b) State **two** functions of the part labeled A. (2 marks)

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c) Name the **two** compounds or substances transported by the tissue from the leaf to other parts of the plant. (2 marks)

d) Explain the role of xylem in plant nutrition. (2 marks)

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5. The results in the table below shows the effects of some conditions on seed germination. In each experiment, all other environmental conditions were kept constant except for the one being investigated.

Experiment	Treatment	% Germination
I	Seeds placed in a tightly closed container with pyrogallic acid.	0
II	(i) Seeds kept in source of light. (ii) Seeds kept in dark cupboard.	96 97
III	(i) Seeds kept in refrigerator 4 ⁰ C. (ii) Seeds kept in oven 60 ⁰ C. (iii)Seeds kept at 35 ⁰ C.	0.5 0 92
IV	Dry seeds in closed container. Moist seeds in a closed container.	0 87

a) (i) What was the purpose of pyrogallic acid in experiment (i) (1 mark)

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(ii) State the aim of the experiment (ii) (1 mark)

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b) (i) Account for the results obtained in experiment set-up (iii) (3 marks)

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(ii) State why 100% germination was not achieved in experiment (ii) and (iv). **(1 mark)**

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c) Of what biological significance is the condition necessary for germination been investigated by experiment (iv)? **(2 marks)**

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SECTION B – 40 MARKS

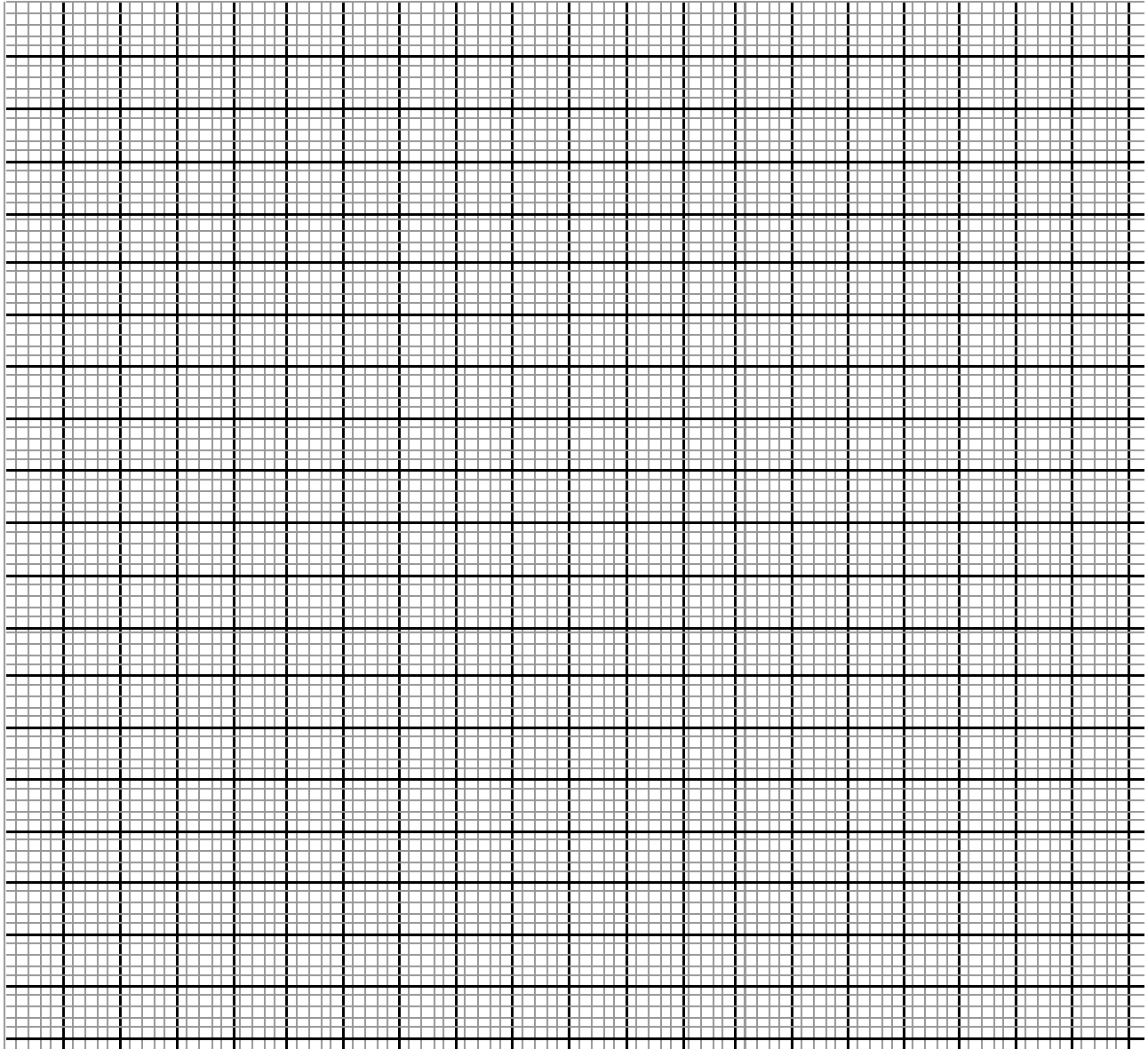
Answer Question 6(Compulsory)

6. (a) An experiment was carried out to investigate the population of a certain micro-organism. Two petri dishes were used. Into the petri dish labeled M 30cm³ of a culture medium was placed while 30cm³ of the same culture was placed in petri dish labeled N. Equal numbers of micro-organisms were introduced in both petri dishes. The set-ups were then incubated at 35⁰C. The number of micro-organisms in each petri dish were determined at equal intervals for a period of 60 hours. The results were as shown in the table below.

Relative number of micro-organisms	M	40	40	180	280	1200	1720	1600	1840	1560	600
	N	40	40	120	200	680	560	560	600	600	400

Time in hours		0	5	10	7.	23	30	35	42	45	60
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- a) (i) On the same axes draw the graphs of relative number of micro-organisms against time on the grid provided.



- (ii) After how many hours was the difference between the two populations greatest? (1 mark)

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- (iii) Work out the difference between the two populations at 50 hours

(2 marks)

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(iv) With a reason state the effect on the population of micro-organisms in petri dish M if the temperature was raised to 60°C after 20 **hours**

(2 marks)

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(v) Account for the shape of the curve for population in petri dish N between 46 and 59 hours

(3 marks)

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b) Explain how the osmotic pressure in the human blood is maintained at normal level **(5 marks)**

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7. (a) Describe how budding takes place in **yeast**
(5 marks)

(b) Discuss the economic importance of **bacteria**
(10 marks)

(c) Describe the methods of excretion in **plants**
(5 marks)

8. (a) Describe the process of photosynthesis in green **plant**
(10 marks)

(b) Explain the various conditions necessary for germination
(10 marks)

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For Marking Schemes Contact 0746 222 000 / 0742 999 000